KAIZEN MATHS

# Lesson starter

**Matrices: Operations, Determinants and Inverses**

Algebraic Matrices

Name _____

**S1**

The matrix $\begin{bmatrix} 1 & 4 \\ -5 & x \end{bmatrix}$ is singular. Find x.

**S2**

Given $\det \begin{bmatrix} 5 & -6 \\ -4 & x \end{bmatrix} = 6$, find x.

**S3**

Solve for X: $X + \begin{bmatrix} -5 & 4 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -8 & -3 \end{bmatrix}$.

**S4**

Given $\begin{bmatrix} 4x & 3 \\ -1 & 6 \end{bmatrix} = \begin{bmatrix} 16 & 3 \\ -1 & 6 \end{bmatrix}$, find x.



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# Exit ticket

**Matrices: Operations, Determinants and Inverses**

Algebraic Matrices

Name _____

**E1**

Given x > 0 and $\begin{bmatrix} x^2-2 & y-x \\ 2z+y & z+x \end{bmatrix} = \begin{bmatrix} 14 & -7 \\ 7 & 9 \end{bmatrix}$, find x, y and z, then evaluate $x^2 + 2y - z$.

**E2**

The matrix $\begin{bmatrix} 1 & 2 \\ -5 & x \end{bmatrix}$ is singular. Find x.

**E3**

Given $\det \begin{bmatrix} 1 & -4 \\ -1 & x \end{bmatrix} = 8$, find x.



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# Teacher answers & working

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**S1**

The matrix $\begin{bmatrix} 1 & 4 \\ -5 & x \end{bmatrix}$ is singular. Find x.

**Answer:**

-20

**Working:**

A singular matrix has determinant 0.

Use det(A) = ad − bc for a 2 by 2 matrix.

$\det(A) = x - (4)(-5)$.

Set the determinant equal to 0: x − -20 = 0.

Move the constant term to the other side: x = -20.

Divide by 1.

$x = -20$.

**S2**

Given det $\begin{bmatrix} 5 & -6 \\ -4 & x \end{bmatrix}$ = 6, find x.

**Answer:**

6

**Working:**

Use $\det(A) = ad - bc$.

$\det(A) = 5x - (-6)(-4)$.

Set equal to 6: 5x − 24 = 6.

Solve to get x = 6.

**S3**

Solve for X: X + $\begin{bmatrix} -5 & 4 \\ -4 & 1 \end{bmatrix}$ = $\begin{bmatrix} 1 & -1 \\ -8 & -3 \end{bmatrix}$.

**Answer:**

$\begin{bmatrix} 6 & -5 \\ -4 & -4 \end{bmatrix}$

**Working:**

Subtract the same matrix from both sides.

$X = \begin{bmatrix} 1 & -1 \\ -8 & -3 \end{bmatrix} - \begin{bmatrix} -5 & 4 \\ -4 & 1 \end{bmatrix}$.

Subtract corresponding entries.

$X = \begin{bmatrix} 6 & -5 \\ -4 & -4 \end{bmatrix}$.

# Teacher answers & working

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**S4**

Given $\begin{bmatrix} 4x & 3 \\ -1 & 6 \end{bmatrix} = \begin{bmatrix} 16 & 3 \\ -1 & 6 \end{bmatrix}$, find x.

**Answer:**

4

**Working:**

Equal matrices have equal corresponding entries.

Check the fixed entries first: the top-right, bottom-left, and bottom-right entries already match.

Compare the top-left entries: 4x = 16.

Divide by 4.

$x = 4$.

**E1**

Given x > 0 and $\begin{bmatrix} x^2 - 2 & y - x \\ 2z + y & z + x \end{bmatrix} = \begin{bmatrix} 14 & -7 \\ 7 & 9 \end{bmatrix}$, find x, y and z, then evaluate $x^2 + 2y - z$.

**Answer:**

x = 4, y = -3, z = 5; expression value = 5

**Working:**

Equal matrices have equal corresponding entries.

Compare the top-left entries: $x^2 - 2 = 14$.

Add 2: $x^2 = 16$.

Since x > 0, x = 4.

Compare the top-right entries: y - x = -7.

Substitute x = 4: y - 4 = -7, so y = -3.

Compare the bottom-left entries: 2z + y = 7.

Substitute y = -3: 2z + (-3) = 7.

So $2z = 10$, and $z = 5$.

Check the bottom-right entry: z + x = 5 + 4 = 9.

Evaluate: $x^2 + 2y - z = (4)^2 + 2(-3) - 5 = 5$.

# Teacher answers & working

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**E2**

The matrix $\begin{bmatrix} 1 & 2 \\ -5 & x \end{bmatrix}$ is singular. Find x.

**Answer:**

-10

**Working:**

A singular matrix has determinant 0.

Use det(A) = ad − bc for a 2 by 2 matrix.

$\det(A) = x - (2)(-5)$.

Set the determinant equal to 0: x − -10 = 0.

Move the constant term to the other side: x = -10.

Divide by 1.

$x = -10$.

**E3**

Given det $\begin{bmatrix} 1 & -4 \\ -1 & x \end{bmatrix}$ = 8, find x.

**Answer:**

12

**Working:**

Use $\det(A) = ad - bc$.

$\det(A) = x - (-4)(-1)$.

Set equal to 8: x − 4 = 8.

Solve to get x = 12.